

Graphene-assisted electromagnetically induced transparency-like terahertz bio-sensor for ultrasensitive detection using optics-enhanced sensing

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Abstract: In recent years, biosensors based on terahertz metamaterials have emerged as a prominent area of research. However, studies focusing on enhancing biosensing performance through optical pumping to improve sensitivity are scarce. This paper proposes a graphene-based electromagnetically induced transparency-like (EIT-like) optics-enhanced terahertz metamaterials biosensor. Using this biosensing platform, we successfully detected glycine through amplitude modulation, achieving a limit of detection (LOD) of 1 pg/mL. Additionally, we developed an optics-enhanced sensing method where the conductivity of graphene is reduced, resulting in a significant increase in the amplitude of the device's transmission spectrum as optics-enhanced. Experimental results demonstrate that the change in the transmission amplitude at varying concentrations exhibits a quasi-linear relationship as the optical-energy-density is varied from 0 $\mu\text{J}/\text{cm}^2$ to 119.7 $\mu\text{J}/\text{cm}^2$. For a low concentration of glycine (32 pg/mL), the optics-enhanced sensing peak amplitude rate of change (AR) reached a maximum of 857.5% at an optical-energy-density of 119.7 $\mu\text{J}/\text{cm}^2$. Furthermore, the graphene-based terahertz metasurfaces biosensor (Gra + Ms) successfully detected a glycine concentration of 1 pg/mL, which would have been undetectable in the absence of optics. These findings suggest that the sensitivity of the Gra + Ms sensor is significantly enhanced under the optical pumping, allowing for more precise detection of low glycine concentrations. This study offers a promising new direction for the practical application of terahertz metamaterials in biosensing.

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1. Introduction

Terahertz (THz) waves are electromagnetic waves with frequency ranging from 0.1-10 THz [1], positioned between infrared and microwave frequencies. They occupy a unique position at the junction of electronics and photonics, exhibiting distinctive electromagnetic properties such as low-energy single pulses [2], coherence [3–5], and high transmittance [6,7]. THz is located in a band that happens to be in the characteristic vibrational band of many biomolecules such as protein molecules, fungi, bacteria, cells, etc. These biomolecules have a strong resonance absorption of THz waves, which allows for the detection of biomolecules. Moreover, due to their extremely low photon energy, THz waves cause minimal disruption to the structure of biomolecules, making them significantly less destructive than conventional detection methods. Metamaterial [8,9] refer to a novel class of artificially engineered composite electromagnetic materials that exhibit extraordinary physical properties not found in natural substances. When

coupled with THz waves, metamaterials can generate a strong electromagnetic response [10,11]. Furthermore, they exhibit characteristics such as localized electric field enhancement and high sensitivity to surrounding dielectric properties. As a result, THz metamaterial sensors have garnered significant attention from the global research community. Currently, THz metamaterial sensors are widely used in various fields, including medical diagnosis [12–16], biomolecular detection [17–20], food and drug safety [21,22] on.

Glycine is a non-essential amino acid commonly used as a food additive. Excessive intake of glycine can lead to nutritional imbalances, potentially impacting overall health. Additionally, glycine serves as a marker for gliomas, as elevated levels are associated with accelerated tumor proliferation and progression [23]. Therefore, achieving trace detection of glycine is crucial for human health and clinical diagnosis. Currently, the primary methods for the detection of glycine include liquid chromatography [24,25] and liquid chromatography-mass spectrometry [26,27], but these techniques suffer from complicated procedures, long detection times, and low sensitivity. The employment of THz metamaterials for ultrasensitive, nondestructive glycine detection could significantly enhance medical testing, food safety, and other applications. While many studies have explored THz metamaterials for amino acid detection, reports specifically focused on glycine are limited. In 2021, Zhang et al. [28] proposed has been THz chiral metamaterial sensor for amino acid solution with a LOD of 0.01 mg/mL. In 2022, Ma et al. [29] demonstrated a THz metamaterial sensor for the detection of common (inositol, leucine, taurine) used in the food industry with a minimum sample concentration of 0.2 mg/mL for detection. In 2023, Li et al. [30] introduced a hybrid biosensor comprising of a plasma-induced transparent hypersurface and graphene for the sensing and detection of amino acid molecules with a LOD of 10 ng/mL. In the same year, Yao et al. [31] presented a terahertz biosensor using a hybrid hypersurface of graphene/graphite carbon nitride heterojunction for the detection of tyrosine molecules with a detection limit concentration of 3.54 ng/mL.

However, THz metamaterial-based biosensors generally face challenges related to high detection limits for amino acids and low detection sensitivity. Therefore, achieving trace detection with these biosensors has become a significant research focus [32,33]. In recent years, electromagnetically induced transparency effect has been widely used in metasurface due to its advantages, including a narrow transparent window, strong local fields, and low radiation loss [34,35]. The transparent window of the EIT metasurface can change significantly with a minimal perturbation in the dielectric environment [36,37], making it an ideal platform for ultra-sensitive sensors. Currently, extensive research is being conducted on ultrasensitive biomolecule detection using two-dimensional active tunable materials combined with ultra-surfaces, with graphene emerging as a key focus due to its ease of integration, low loss, high carrier mobility, and dynamic tunability. Xu et al. [38] proposed a patterned flexible graphene THz sensor for bio-interfacial pesticide detection with a detection limit concentration of 0.13 ng/mL. Yao et al. [39] introduced a novel THz metamaterial biosensor based on patterned graphene and chalcogenide for the detection of whey proteins with a LOD of 6.2 ng/mL. Zhou et al. [40] utilized graphene-metasurface structure for the detection of doxycycline hydrochloride with a detection limit concentration of 10 μ g/mL. Xu et al. [41] realized the detection of chlorpyrifos fructose and formic acid using a biosensor with a graphene-metamaterial heterostructure platform with a LOD of 100 ng/mL and 20 ng/mL, respectively. However, when relying on changes in the amplitude of the THz transmission spectrum as a sensing indicator, the amplitude variations tend to be minimal at low concentrations, complicating the detection of trace substances. Utilizing the optoelectronic properties of graphene to alter its conductivity through optical pumping holds promise for achieving highly sensitive biomolecular sensing.

In this study, an EIT-like graphene-based THz metasurface biosensor (Gra + Ms) is proposed. By applying different concentrations of glycine onto the surface of the Gra + Ms sensor, we modulated the Fermi level of graphene, which in turn altered the amplitude of the device's

transmission spectrum. This experimental setup achieved an exceptionally low LOD of 1 pg/mL for glycine. Leveraging the optical properties of graphene, we implemented optics-enhanced sensing and explored the variation in transmission peak amplitude under different optical-energy-density levels. Experimental results demonstrated that the transmission peak amplitude increased with higher optical-energy-density, and the change in transmission peak amplitude at varying concentrations showed a quasi-linear trend. By calculating the rate of change in peak amplitude before and after optics-enhanced, we observed a maximum increase of 857.5%, indicating that the sensitivity of the Gra + Ms sensor is significantly enhanced under illumination, making it highly responsive to low concentrations of glycine optics-enhanced. These findings suggest that the sensitivity of the Gra + Ms sensor is significantly enhanced under optics-enhanced, allowing for more precise detection of low glycine concentrations. This study optimizes the performance of the proposed biosensor and lays the groundwork for further research in this field.

2. Results and discussion

2.1. Metasurface fabrication and characterization

The fabrication process of the Gra + Ms THz metamaterial biosensor proposed in this paper is shown in Fig. 1(a). The process begins with spin-coating a 5- μm -thick polyimide (PI) film onto a 300- μm -thick quartz substrate. Then, the 0.2- μm -thick metallic microstructure is formed on the PI film by photolithography, development and magnetron sputtering. Finally, the 1.0 cm $\times$ 1.0 cm monolayer graphene is transferred to the metallic aluminum surface by wet method, and the Gra + Ms device is fabricated. Figure 1(b) shows the optical microscope picture of the periodic structure. Glycine solution was added dropwise to this sensor for biosensing, and its three-dimensional structure is shown schematically in Fig. 1(c).

The structural material was modeled and simulated using the time domain finite element method, with the boundary conditions set as follows, x-axis is the direction of the electric field and y-axis is the direction of the magnetic field. The unit structure of the EIT-like metasurface consists of two split-ring resonators (SRRs), as shown in Fig. 1(d), with the following geometrical parameters, $p = 80\ \mu\text{m}$, $m = 62\ \mu\text{m}$, $n = 30\ \mu\text{m}$, $l = 31\ \mu\text{m}$, $d = 42\ \mu\text{m}$, $w = 9\ \mu\text{m}$ respectively. The thickness of the metal-aluminum film is all 0.2 μm , and its conductivity is $3.56 \times 10^7\ \text{S/m}$. The thickness of the substrate polyimide is 5 μm , its dielectric constant is 3.1, and the corresponding dielectric loss angle is 0.05.

In the test, Raman spectroscopy of the monolayer graphene was performed using a 514 nm laser as shown in Fig. 1(e). The result indicates that the intensity ratio of the 2D peak (about $2698\ \text{cm}^{-1}$) to that of the G peak (about $1578\ \text{cm}^{-1}$) is about 3.3, and the half peak width of the 2D peak is $55\ \text{cm}^{-1}$. The above parameters can prove the excellent quality of this graphene [42].

To test the response of THz waves to the concentration of glycine, five solutions with different concentrations were prepared, $C_1 = 32\ \text{pg/mL}$, $C_2 = 3.33\ \text{ng/mL}$, $C_3 = 0.33\ \mu\text{g/mL}$, $C_4 = 2.33\ \mu\text{g/mL}$, $C_5 = 0.113\ \text{mg/mL}$. All the experimental data presented in this work are averaged after 3 tests.

Device testing was carried out using a THz time-domain spectroscopy (TDS, Advantest TAS7500SU) is shown in Fig. 2. To avoid the possible impact of moisture in the atmosphere, dry air was introduced into the THz-TDS setup to maintain a constant environment. The room temperature is about 25°C ; the room humidity is about 2.4%, respectively. Bandwidth is 0.5THz-7THz, the signal-to-noise ratio is greater than 65 dB, the frequency resolution is 7.6 GHz, and throughput is 200 ms/scan. The normally incident THz wave is along +z, with electric component along the x-direction. The externally pumped light source used in the experiments was a laser with a wavelength of 532 nm and a spot diameter of 3 mm emitter.

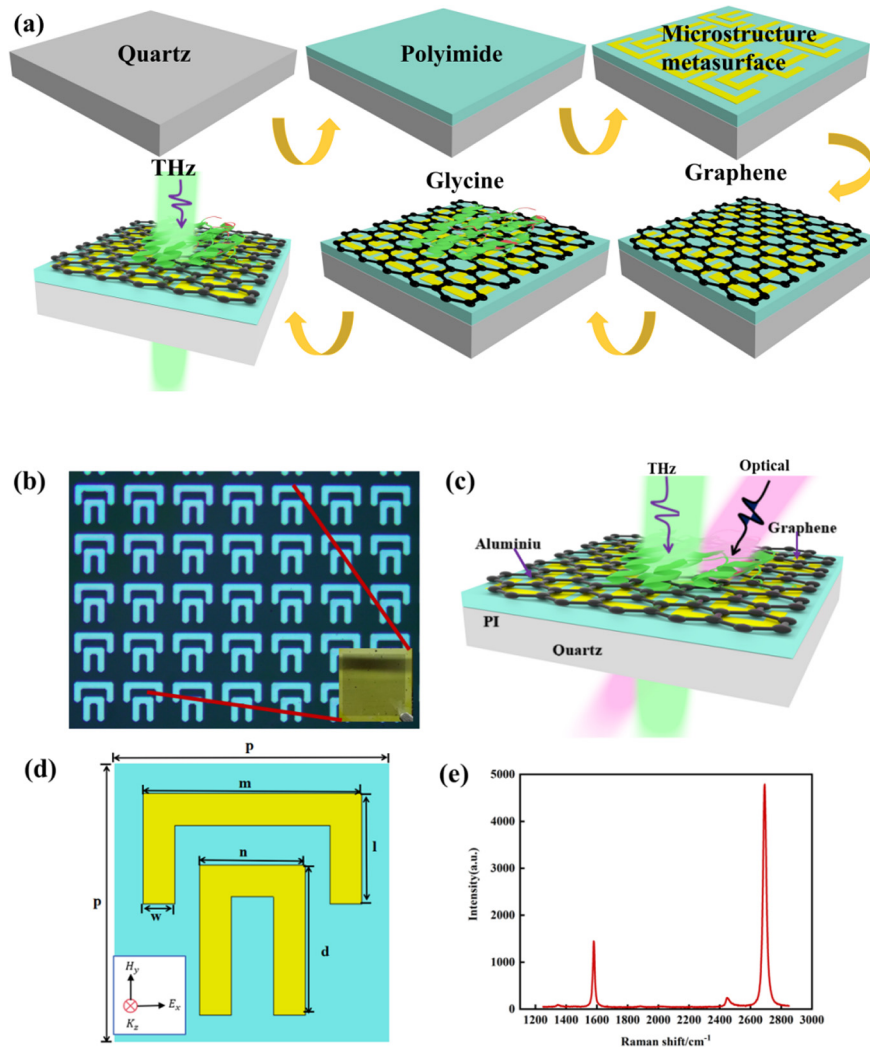


Fig. 1. Manufacture and characterization of the Gra + Ms. (a) Manufacture process of the Gra + Ms. (b) 3D schematic of the Gra + Ms structure. (c) Electron micrograph of the Gra + Ms. (d) Unit cell of the EIT-like metasurface. (e) Raman spectrum of monolayer graphene.

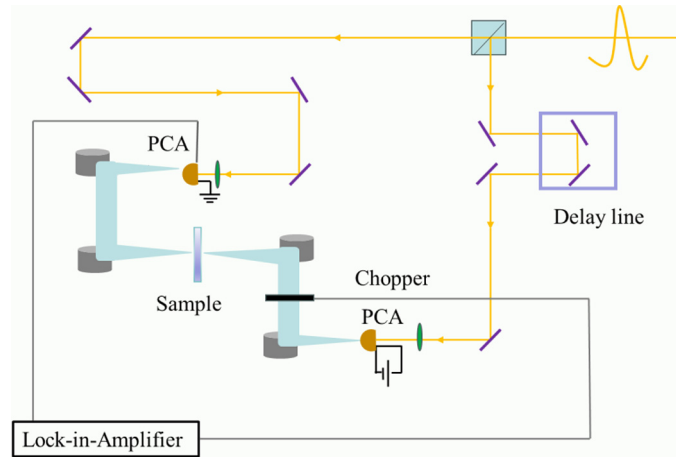


Fig. 2. The schematic view of all-fiber THz-TDs measurement setup.

2.2. Eit-like metasurface mechanism and theoretical model

To investigate the resonance mechanism of the designed Eit-like metasurface, the transmission spectra of the two open-ring resonators SRR_1 (red), SRR_2 (green), and the combined cell structure two SRRs (blue) were simulated, as shown in Fig. 3(a). As shown in the curves, the open-ring resonators SRR_1 (bright model 1) and SRR_2 (bright model 2) have significant resonance, with resonance valleys at 0.96 THz and 1.21 THz, respectively. when the two SRRs are combined (EIT-like metasurface), they couple to produce a resonance peak with a narrow bandwidth, reaching an amplitude of approximately 0.8 at 1.088 THz (EIT-like). Figure 3(b) displays the experimental transmission spectral line (red) alongside the simulated transmission spectral line (green), both showing a W-shaped profile. A comparison reveals that the simulation results align closely with the experimental measurements, although there are slight discrepancies in the intensity of the peaks and valleys. The primary reason for the deviation may be the error caused by the Gra + Ms sample processing. Nevertheless, the simulation results substantiate the accuracy of the metamaterial fabrication and validate the measurement outcomes.

To analyze the generation mechanism of EIT-like effect in the metasurface structure, the electric field distributions at the resonance frequencies of 0.93 THz, 1.088 THz, and 1.22 THz were simulated, as shown in Fig. 3(c)–3(e), respectively. For the open-ring resonators SRR_1 (bright model 1) and SRR_2 (bright model 2), the electromagnetic wave is incident perpendicular to the surface of the metasurface, with the electric field aligned parallel to the bottom of the open resonator ring. In this case, the THz wave polarized along the x-direction excites the double open-ring resonators, producing dipole resonance, leading to two resonance valleys at frequencies f_1 and f_3 . Due to the differing dimensions of the two resonators, the frequency and intensity of the resonance in response to the incident electromagnetic wave vary. From Fig. 3(c)–3(e), it can be seen that at the first valley (f_1) with a frequency of 0.93 THz, the electric field is predominantly concentrated on the open ring SRR_1 (bright model 1), generating the first resonance valley. At 1.22 THz (f_3), the electric field is mainly concentrated on the open ring SRR_2 (bright model 2), creating a similar resonance valley. Conversely, at 1.088 THz (f_2), the two dipole resonances interact and undergo coherent phase cancellation phenomenon, causing the electric field to concentrate between the two open rings and thereby reducing the field strength. Since the close frequencies of f_1 and f_3 , this suppression of both dipole resonances leads to the formation of a narrowband transparent window at f_2 , the EIT-like effect is produced [43].

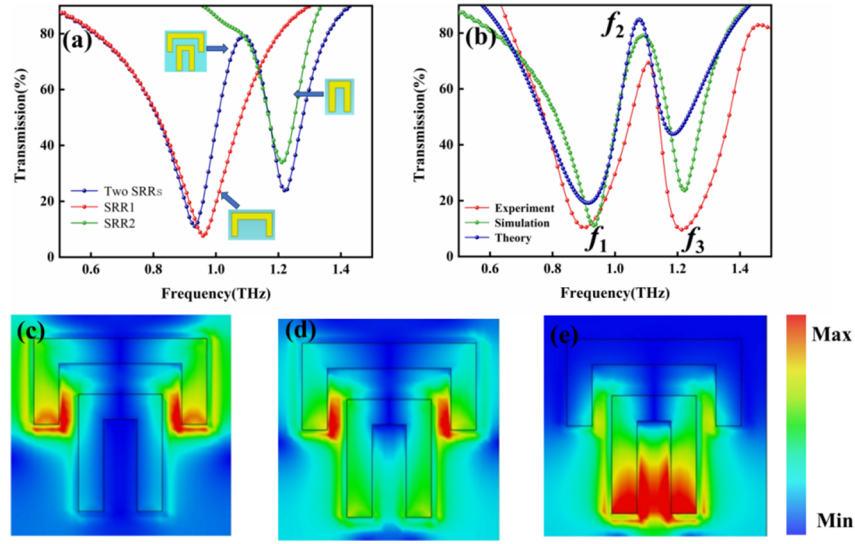


Fig. 3. Transmission spectrum and electric field of the Gra + Ms. (a) Simulated transmission spectra of SRR₁ (red), SRR₂ (green) and two SRRs (blue). (b) Experimental, simulated and theory fitted transmission spectra. (c)-(e) Electric field intensity distribution at f_1 , f_2 and f_3 .

The EIT-like electromagnetic response can be explained by time-domain coupled mode theory (TCMT), grounded in the fundamental principles of optics and electronics. In this context, the metasurface resonator can be modeled as a two-port system, where port 1 represents the incident wave and port 2 represents the transmitted wave. The dynamic equation of resonance amplitude can be written as [44,45]:

$$\frac{d}{dt} \begin{pmatrix} a_b \\ a_d \end{pmatrix} = (j\Omega - \Gamma_e - \Gamma_i) \begin{pmatrix} a_b \\ a_d \end{pmatrix} + D^T \begin{pmatrix} s_{1+} \\ s_{2+} \end{pmatrix}, \quad (1)$$

$$\begin{pmatrix} s_{1-} \\ s_{2-} \end{pmatrix} = C \begin{pmatrix} s_{1+} \\ s_{2+} \end{pmatrix} + D \begin{pmatrix} a_b \\ a_d \end{pmatrix}. \quad (2)$$

The meaning of denotations in the above formulas is as follows: a_b and a_d are the normalized resonance amplitude of bright model 1 and bright model 2 resonators, respectively; s_{1+} and s_{2+} (or s_{1-} and s_{2-}) are the normalized amplitude of incoming (or outgoing) waves at port 1 and port 2, respectively. When the resonant system is excited by a monochromatic wave, the scattering matrix of the EIT structure can be expressed as follows:

$$S = C + D[j\omega I - (j\Omega - \Gamma_e - \Gamma_i)]^{-1} D^T. \quad (3)$$

Here, $C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} j\sqrt{\Gamma_{e1}} & 0 \\ j\sqrt{\Gamma_{e1}} & 1 \end{pmatrix}$, $\Omega = \begin{pmatrix} \omega_1 & -\kappa \\ -\kappa & \omega_2 \end{pmatrix}$, $\Gamma_e = \begin{pmatrix} \Gamma_e & 0 \\ 0 & 0 \end{pmatrix}$, $\Gamma_i = \begin{pmatrix} \Gamma_{i1} & 0 \\ 0 & \Gamma_{i2} \end{pmatrix}$.

The matrix C is the direct scattering matrix of the system in the absence of the resonator, and the matrix D represents the coupling between the resonant modes and the incident and transmitted

waves, and the simplified transmission curve can be expressed as [46],

$$S_{21} = \frac{(j\omega - j\omega_1 + \Gamma_{i1})(j\omega - j\omega_2 + \Gamma_{i2}) + \kappa_2}{(j\omega - j\omega_1 + \Gamma_{i1} + \Gamma_{e1})(j\omega - j\omega_2 + \Gamma_{i2}) + \kappa_2}, \quad (4)$$

where ω_1 and ω_2 represent the resonant frequencies of the two modes, respectively; Γ_{e1} and Γ_{i1} , Γ_{i2} represent the radiative and non-radiative attenuation of the two modes, respectively; and κ represents the coupling coefficient between the two modes.

The simulated transmission spectrum was fitted by applying Eq. (4). The transmission spectrum of the EIT-like structure predicted by the TCMT was shown as the blue spectral line in Fig. 3(b), from which it can be seen that the theoretical fitting results (TCMT) are in good agreement with the simulation results (EIT-like). Due to the higher Q value of the transmission peak, this metasurface has higher sensitivity and is ideal for biosensors.

2.3. Sensing performance of the Gra + Ms biosensors

To investigate the sensing performance of the Gra + Ms biosensor, different concentrations of glycine were dropped onto the surface of the sensor, and experiments were conducted after allowing the solution to evaporate. Additionally, to emphasize that the ultrasensitive detection capabilities of the Gra + Ms sensor primarily arise from graphene, we examined the sensing performance of the metasurface structured Ms-only device under the same experimental conditions. The transmission spectra of the metasurface structured Ms-only device detecting varying glycine concentrations are presented in Fig. 4(a). The transmission spectra of the metasurface structured Gra + Ms device detecting varying glycine concentrations are presented in Fig. 4(b). For ease of analysis, the transmission valleys and peak in the spectra are designated as f_1 , f_2 and f_3 . From Fig. 4(a), it is evident that the overall transmission spectral profile remains largely unchanged with varying glycine concentrations, showing only minor variations at f_2 . In contrast, the results in Fig. 4(b) demonstrate that as glycine concentration increases, the transmission spectrum exhibits significant changes due to variations in the conductivity of graphene. Notably, the amplitudes of the transmission valleys f_1 and f_3 decrease, while the transmission peak f_2 increases continuously. The above phenomena indicate that the amplitude changes at frequencies f_1 , f_2 and f_3 can all be used for glycine concentration detection, further supporting the capability of the Gra + Ms biosensors for the ultrasensitive glycine detection.

To quantitatively assess the sensing performance, the parameter $\Delta T = T_{Ci} - T_0$ is defined, where T_{Ci} (T_0) is the amplitude of the transmission peaks (valleys) in the presence (absence) of glycine. The curve of ΔT with the glycine concentration, corresponding to Fig. 4(a) is shown in Fig. 4(c). The results show that only the value of ΔT for f_2 varies slightly at concentrations C_2 and C_3 , making it challenging for the Ms sensor to achieve ultrasensitive glycine detection. Figure 4(d) demonstrates the ΔT curves for the Gra + Ms devices under varying glycine concentration. The results demonstrate that with glycine concentration increases, the changes in ΔT at the three frequency points become more pronounced, showing a good linear correlation. Both f_1 and f_3 exhibit a decreasing trend with increasing glycine concentration. Specifically for the frequency f_1 , $|\Delta T|$ changes from 2.372% to 9.67% as the concentration vary from C_1 to C_5 . Similarly, for frequency f_3 , $|\Delta T|$ changes from 6.923% to 12.94% over the same concentration range. Conversely, the ΔT for f_2 (purple) increase with glycine concentration, when $C_1 = 32$ pg/mL, $\Delta T = 1.61\%$; when $C_2 = 3.33$ ng/mL, $\Delta T = 3.25\%$; and when the concentration increased to $C_5 = 0.113$ mg/mL, ΔT reached a maximum of 9.72%.

To further clarify the sensing effect at the three frequency points, we fitted the linear slope values of the relationship curves for ΔT against glycine concentration at frequencies f_1 , f_2 and f_3 , which are displayed in Fig. 4(d). It can be found that the slope corresponding to f_2 is the largest and the slope for f_3 is the smallest, indicating that f_2 serves as a reliable indicator point for the sensor's overall performance.

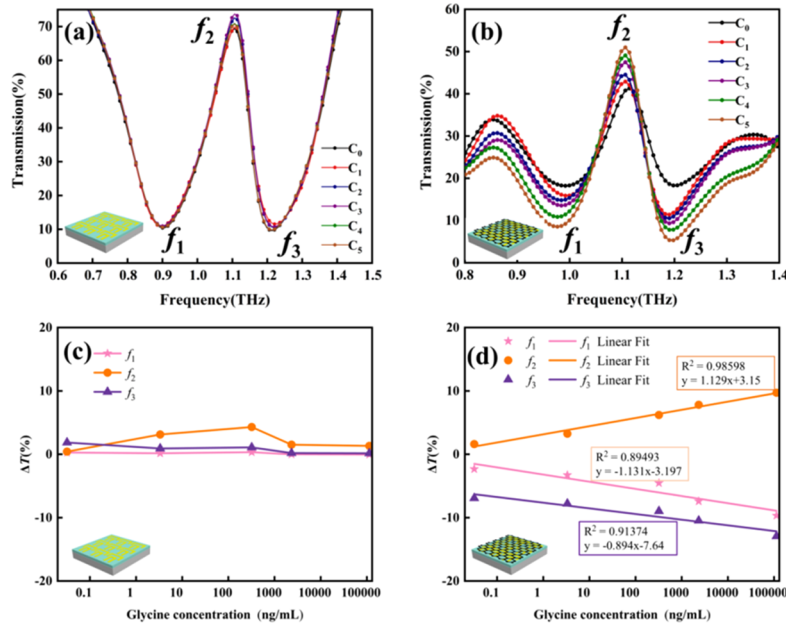


Fig. 4. Sensing performance of the biosensors. (a) and (b) Transmission spectrum of the Ms (Gra + Ms) at different glycine concentrations. (c) and (d) The curve of ΔT of the Ms (Gra + Ms) with glycine concentration ($C_1 = 32$ pg/mL, $C_2 = 3.33$ ng/mL, $C_3 = 0.33$ μ g/mL, $C_4 = 2.33$ μ g/mL, $C_5 = 0.113$ mg/mL).

The internal mechanism underlying the change in transmission spectrum amplitude under glycine concentrations can be explained by the change of Fermi level of graphene, illustrated in Fig. 5. For intrinsic graphene, the Fermi level is located at the Dirac point, where both carrier density and conductivity are zero. The graphene used in the experiment is *p*-type doped, with its initial Fermi level (E_{F0}) is located in the valence band near the Dirac point, as shown in I of Fig. 5. When a glycine concentration of 32 pg/mL (C_1) is introduced, glycine molecules dope the graphene, shifting the Fermi level from E_{F0} to E_{F1} (II in Fig. 5). This doping reduces the surface conductivity of graphene, decreasing the loss of incident THz waves and consequently increasing the amplitude of the transmission spectrum. The amplitude of f_2 remains relatively stable at low glycine concentrations. As glycine concentration increases, the Fermi level of graphene continues to rise from E_{F1} to E_{F2} (III in Fig. 5), approaching the Dirac point. The upward shift enhances the interaction between graphene and the molecules, strengthening the coupling of biomolecules to the electric field of the terahertz waves. When a concentration of 0.113 mg/mL, the coupling intensifies further, resulting in the maximum amplitude of the transmission spectrum at f_2 . Moreover, when the Fermi level of graphene is close to the Dirac point, the electric field coupling becomes so pronounced that the transmission valleys at f_1 and f_3 reach their minimum values.

2.4. Optics-enhanced biosensing

In order to further improve the sensitivity of sensor concentration detection, this study leverages the tunable property of graphene conductivity to enhance the interaction between biomolecules and graphene under applied optics excitation. This enhancement leads to ultrasensitive detection capabilities in the biosensor. In this paper, we investigated the THz transmission spectra of four different glycine concentrations (concentrations of 32pg/mL (C_1), 3.33 ng/mL (C_2), 0.33 μ g/mL (C_3), and 2.33 μ g/mL (C_4), respectively) by applying 532 nm optical pumping to the Gra + Ms

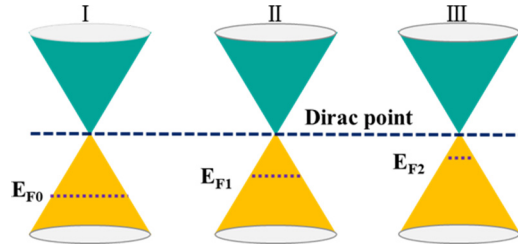


Fig. 5. Schematic diagram of the Fermi level change of graphene at different glycine concentrations.

sensor. The optical-energy-density was varied from $0 \mu\text{J}/\text{cm}^2$ to $119.7 \mu\text{J}/\text{cm}^2$, as shown in Fig. 6. It can be observed from Fig. 6(a)–(d) that the transmission spectra display a consistent pattern, with the transmission peak amplitude at f_2 increasing as the optical-energy-density increases. These findings indicate that the optics excitation enhances the ultrasensitive detection capabilities of the Gra + Ms biosensor, particularly through amplitude changes.

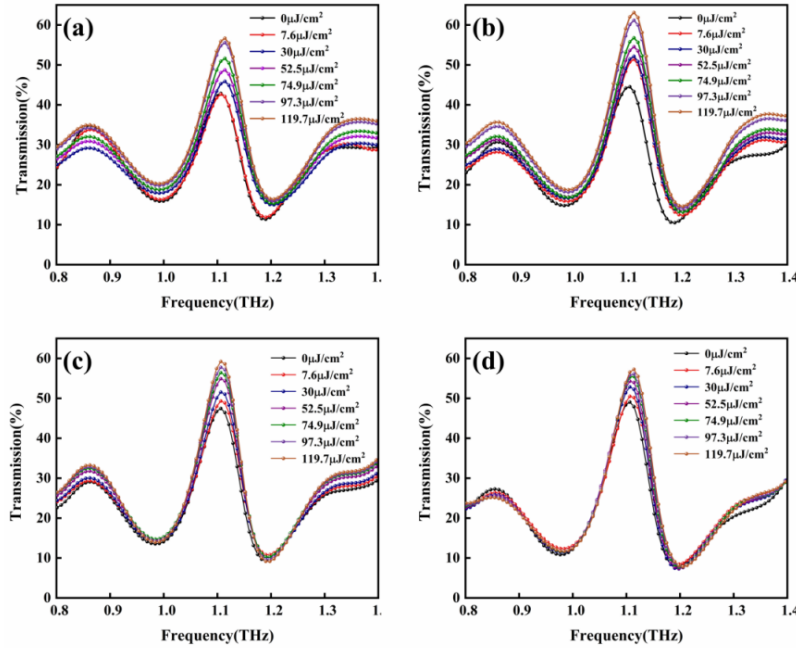


Fig. 6. The THz transmission spectra of four different glycine concentrations under optics excitation. (a)–(d) The glycine concentrations were 32pg/ml (C_1), 3.33 ng/ml (C_2), 0.33 $\mu\text{g}/\text{ml}$ (C_3), 2.33 $\mu\text{g}/\text{ml}$ (C_4), respectively.

To quantify the effect of optics-enhanced sensing, we define here $\Delta T = T_{ED} - T_0$, $\Delta T_0 = T_{C_i} - T_0$, $AR = \frac{\Delta T - \Delta T_0}{\Delta T_0} \times 100\%$ (the rate of change of transmission amplitude before and after light-excited sensing), where T_{ED} represents the amplitude of the f_2 transmission peak in the presence of optical-energy-density, T_0 is the amplitude of the f_2 transmission peak in the absence of optical-energy-density without glycine, and T_{C_i} is the amplitude of the transmission peak in the absence of optical-energy-density with glycine. The quantitative analysis of optics-enhanced sensing is shown in Fig. 7. Figure 7(a) shows the change in the peak amplitude of f_2 transmission (ΔT) under different optics excitations, while Fig. 7(b) demonstrates the rate of change in the peak

amplitude (AR) before and after optics-enhanced sensing. The results indicate that the ΔT curves for the four different concentrations of glycine exhibit a nearly linear with optical-energy-density (ED). These findings demonstrate that the rate of change in the peak amplitude increases with higher ED . When the ED was varied from 0 $\mu\text{J}/\text{cm}^2$ to 119.7 $\mu\text{J}/\text{cm}^2$, the maximum variation (ΔT) of the transmission peak amplitude for four glycine concentrations 32 pg/mL , 3.33 ng/mL , 0.33 $\mu\text{g}/\text{mL}$, and 2.33 $\mu\text{g}/\text{mL}$ reach 15.4%, 21.9%, 17.6%, and 24.8%. The corresponding AR values reached 857.5%, 571.8%, 183.6%, and 61.7%. Notably, for a low concentration of glycine (32 pg/mL), the peak amplitude change reached 185.1% at ED of 30 $\mu\text{J}/\text{cm}^2$ and a maximum of 857.5% at 119.7 $\mu\text{J}/\text{cm}^2$, suggesting that this Gra + Ms biosensor can achieve ultra-sensitivity under optics-assisted excitation.

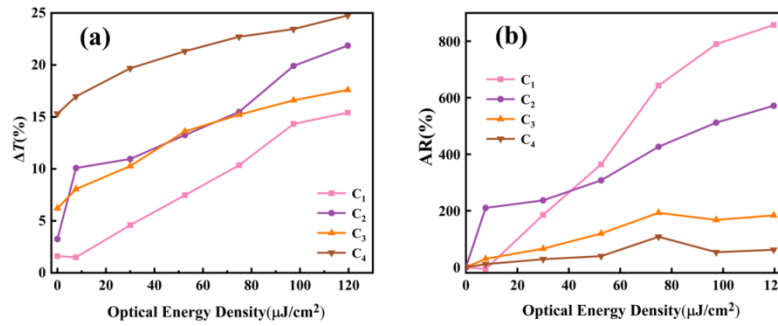


Fig. 7. The quantitative analysis of optics-enhanced sensing. (a) The change in the peak amplitude of f_2 transmission (ΔT) under different optics excitations. (b) The rate of change in the peak amplitude (AR) of f_2 before and after optics-enhanced sensing.

The mechanism of optics-enhanced sensing can also be explained by changes in the Fermi level of graphene. The theoretical conductivity of graphene is modeled as [47–49]:

$$\sigma_{Gr} = \frac{2\sigma^0}{1 + e^{-\frac{\mu}{T}}}, \quad (5)$$

when $\sigma^0 = e^2 T_0 \tau_0 / \pi h^2$ graphene is in the absence of excitation: $T = T_0$ and $\mu = 0$. The graphene conductivity is determined by the carrier densities determined, mainly by the Fermi energy μ and its effective temperature T . Under relatively weak irradiation, $|\mu| \ll T_0, T$, the conductivity of graphene is approximated as:

$$\sigma_{Gr} \approx \sigma_0 \left(1 + \frac{\mu}{2T}\right). \quad (6)$$

When the biosensor is excited by an optics external, for a p-type graphene, optical excitation leads to the increase in carrier temperature, which reduces the chemical potential of graphene, leading to the decrease in conductivity of graphene. According to the Eq. (6), the conductivity of graphene decreases as the optical-energy-density increases, and the μ decrease. Fermi level approaches the Dirac point. Previously, we analyzed that when glycine is added to the biosensor, the Fermi level of graphene remains in the valence band (as shown in Fig. 5). Meanwhile, the effective temperature of the carriers inevitably increases under the continuous optics excitation. As shown in the Eq. (6), the conductivity of graphene decreases with the increasing optical-energy-density, leading to an increase in the transmission values across all frequency peaks. Thus, in the Gra + Ms biosensor, both glycine and laser light interact with graphene, causing the Fermi level to shift upwards again. Graphene is highly sensitive to changes in carrier concentration due to its position within the valence band. Under the influence of external doping, the Fermi level shifts closer to the Dirac point, altering the conductivity of graphene and ultimately resulting in a significant change in the amplitude of the transmission peak. leading to significant changes

in the amplitude of the f_2 transmission peak. As the optical-energy-density increases to $119.7 \mu\text{J}/\text{cm}^2$, the f_2 transmission peak amplitude reaches the maximum value, which further proves the feasibility of the optics-enhanced biosensing method proposed in this paper.

To validate the optics-enhanced sensing sensitivity proposed in this study, we explored the detection of glycine at concentrations lower than 32 pg/mL , which are typically undetectable by the Gra + Ms sensor without optics excitation. The results were compelling, as the lowest detectable concentration was reduced from 32 pg/mL to 1 pg/mL with the optics-assisted, demonstrating the achievement of ultra-sensitive detection. Figure 8(a) presents the transmission spectrum of the Gra + Ms biosensor with glycine concentration of 1 pg/mL and without glycine

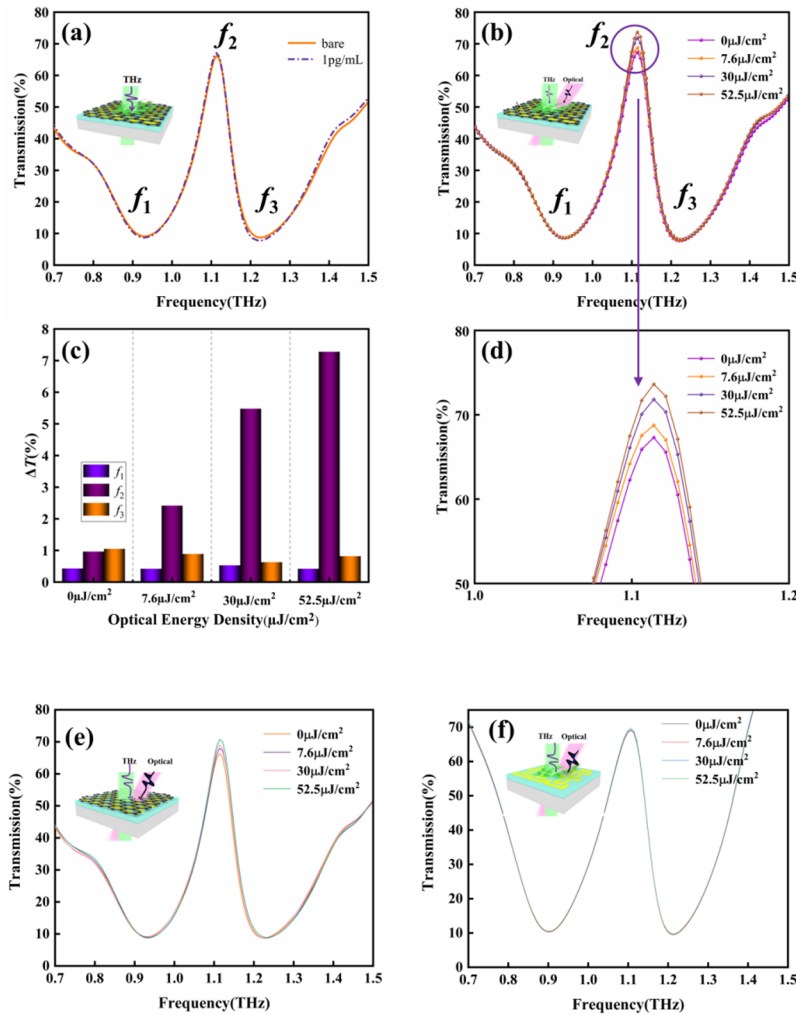


Fig. 8. Quantitative analysis of optics-enhanced low concentration sensing. (a) Transmission spectra with dropwise addition of glycine at a concentration of 1 pg/mL and without glycine under zero optical-energy-density conditions. (b) Transmission spectra with optical-energy-density at a glycine concentration of 1 pg/mL . (c) The amount of variation in peak amplitude (ΔT) at frequencies f_1 , f_2 and f_3 for different optical-energy-density. (d) The transmission peak (f_2) in (b) is magnified in the picture. (e) Gra + Ms sensor with optics excitation and no amino acids. (f) Ms sensor with optics excitation and 1 pg/mL amino acids.

when the optical-energy-density is zero. The results show that the two transmission spectra nearly overlap, with no significant changes indicating that it is difficult for the sensor to detect glycine at such a low concentration without the applying of optics. However, Fig. 8(b) shows the transmission spectra under varying optical-energy-density levels, applied to the Gra + Ms biosensor with 1 pg/mL glycine. The results clearly indicate a significant change in the amplitude of the transmission peak (f_2), as highlighted in Fig. 8(b). This confirms that the Gra + Ms sensor can enhance sensitivity under optics excitation, enabling the detection of glycine at concentrations as low as 1 pg/mL. To quantitatively analyze the sensitivity of optics-enhanced sensing, we conducted a comparative assessment of the transmission peak and valley amplitudes of under varying optical-energy-density conditions compared to those without optical-energy-density and in the absence of glycine. The variation in amplitudes for frequencies f_1 , f_2 , and f_3 , across different levels of ED , illustrated in Fig. 8(c).

From the analysis, it is evident that as the optical-energy-density increased from 0 $\mu\text{J}/\text{cm}^2$ to 52.5 $\mu\text{J}/\text{cm}^2$, the amplitude of the transmission valleys (f_1, f_3) showed minimal change. However, the amplitude of the transmission peak (f_2) showed the significant change, rising from 0.963% to 7.281%. This demonstrates that the Gra + Ms biosensor successfully detected glycine at a concentration of 1 pg/mL under photoexcitation. This result lowered the detection limit of the Gra + Ms biosensor to 1 pg/mL, enhancing the sensor's sensitivity.

Furthermore, to illustrate the optics-enhanced sensing principle of the proposed biosensor, transmission spectra for two control groups were tested: Gra + Ms sensor with optics excitation and no amino acids, and Ms sensor with optics excitation and 1 pg/mL amino acids, as shown in Fig. 8(e) and Fig. 8(f), respectively.

As observed in Fig. 8(e) and Fig. 8(f), the transmission amplitude of f_2 for Gra + Ms sensor varies with changes in optical-energy-density, whereas for Ms sensor, there is no significant change. These results confirm that the ultra-sensitive detection achieved by our proposed biosensor is primarily due to the change in graphene's conductivity induced by optics excitation. Additionally, the presence of amino acid molecules leads to molecular doping of graphene, causing changes in its conductivity. As shown in previous studies, different amino acid concentrations affect the electrical properties of graphene differently. Therefore, the combined effect of light and amino acid molecules alters graphene's conductivity, resulting in changes in the transmission amplitude and enabling ultra-sensitive detection.

Furthermore, it confirms that the Electromagnetically Induced Transparency transmission peaks have highly sensitive properties for biosensing applications. This confirms that the Gra + Ms sensor can enhance sensitivity under optics excitation, enabling the detection of glycine at concentrations as low as 1 pg/mL.

Table 1. Comparison with previous similar work based on THz biosensors^a

Sensors	Analytes	LOD	Reference
Ms	AA	0.01 mg/mL	[21]
Ms	AA	0.2 mg/mL	[22]
Gra + Ms	AA	10 ng/mL	[23]
Gra + GCN	Tyr	3.54 ng/mL	[24]
Gra + PI	CMM	0.13 ng/mL	[31]
Gra + Ms	CMM	20 ng/mL	[34]
Gra + Ms	Gly	1pg/mL	This Work

^aLOD: limit of detection; Gra: Graphene; Ms: metasurfaces; AA: amino acid; CMM: chlorpyrifos methyl molecules; DHC: doxycycline hydrochloride Gly: Glycine; Tyr: Tyrosine

Compared to previous studies (Table 1), the biosensor proposed in this work demonstrates a significant improvement in sensitivity, highlighting the effectiveness of optics-assisted detection in achieving ultra-sensitive performance.

3. Conclusions

In summary, we propose a graphene-based EIT-like THz metamaterial biosensor (Gra + Ms), and analyze the excitation mechanism generating EIT-like resonance peaks. The response is theoretically explained using time-domain coupled mode theory (TCMT). The Gra + Ms biosensor was employed as a detection platform for glycine, with amplitude change serving as the sensing indicator, achieving a LOD as low as 1 pg/mL. Additionally, we demonstrated optics-enhanced sensing, where experimentally results showed a significant increase in amplitude of transmission spectrum with increasing optical-energy-density of optical pumping. The change in transmission peak amplitude (ΔT) varied quasi-linearly across different concentrations of glycine. Specifically, when the ED was varied from 0 $\mu\text{J}/\text{cm}^2$ to 119.7 $\mu\text{J}/\text{cm}^2$ for glycine concentrations of 32 pg/mL, 3.33 ng/mL, 0.33 $\mu\text{g}/\text{mL}$, and 2.33 $\mu\text{g}/\text{mL}$, the maximum reach 15.4%, 21.9%, 17.6%, and 24.8%, respectively. The rate of change (AR) for the optics-enhanced sensing reached 857.5%, 571.8%, 183.6%, and 61.7%, respectively, indicating a substantial enhancement in sensitivity under optics-assisted conditions. Compared with previous similar work based on THz biosensor, we propose a graphene-based EIT-like optics-enhanced THz metamaterial biosensor (Gra + Ms) has high sensitivity and low detection limit and achieves highly sensitive biomolecular sensing. This work holds significant promise for the practical application of terahertz metasurfaces in biosensing, enabling ultra-sensitive detection through optics-assisted techniques.

Funding. National Natural Science Foundation of China (62201496, 61701434, 62375235, 62471431); Special Funding of the Taishan Scholar Project (tsqn201909150); Natural Science Foundation of Shandong Province (ZR2020FK008, ZR202102180769, ZR2022QF054, ZR2021MF014); Qingchuang Science and Technology Plan of Shandong Universities (2023KJ283).

Disclosures. The authors declare that there are no conflicts of interest related to this article.

Data availability. Data underlying the results presented in this paper are not publicly available at this time, but may be obtained from the authors upon reasonable request.

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